

ON COUPLING LEMMA AND STOCHASTIC PROPERTIES WITH UNBOUNDED OBSERVABLES FOR 1-D EXPANDING MAPS

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OUTLINE

- 1 STATISTICAL PROPERTIES
- 2 GENERAL ASSUMPTIONS
- 3 STATEMENT OF RESULTS
- 4 COUPLING IDEAS IN THE PROOF

PHYSICAL MEASURES AND ACIP

- Let $T : M = [0, 1] \rightarrow M$ be a Borel measurable map, which preserves an ergodic probability measure μ .
- By Birkhoff Ergodic Theorem, $\mu(\mathcal{B}_\mu) = 1$, where

$$\mathcal{B}_\mu := \left\{ x : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x) = \int f d\mu, \forall f \in C(M) \right\}.$$

- μ is called a **physical** measure if $Leb(\mathcal{B}_\mu) > 0$.
- If μ is an **ACIP** (absolutely continuous invariant probability), i.e., $\mu \ll Leb$, then μ is physical.

GENERAL QUESTIONS ON ERGODIC THEORY AND STATISTICAL PROPERTIES

- (1) Under what conditions, T admits a unique ergodic ACIP μ ?
- (2) Is (T, μ) mixing, i.e., $\lim_{n \rightarrow \infty} C_n(f, g) = 0$ for any L^2 observables f and g ? Here

$$C_n(f, g) := \left| \int f \cdot g \circ T^n d\mu - \int f d\mu \int g d\mu \right|.$$

Do the correlations decay exponentially fast for smooth observables?

- (3) Does the CLT (central limit theorem) hold? i.e.,

$$\frac{(f + f \circ T + \dots + f \circ T^{n-1}) - n \int f d\mu}{\sqrt{n}} \rightarrow N(0, \sigma^2) \text{ in distribution.}$$

METHODS AND RESULTS

- Transfer operators:

- ▶ Results: Uniformly expanding/hyperbolic map with Big Image Property (BIP), in the 1-d case, it means that

$$\inf\{|TW| : W \text{ is a maximal inverse branch}\} > 0.$$

- ▶ Experts: Ruelle, Henoin, Keller, Liverani, Baladi, Dolgopyat, Hu, Tsujii, Demers, Zhang, etc.

- Markov partition or Young tower:

- ▶ Nonuniformly hyperbolic systems with integrable return times (where the induced system has BIP);
- ▶ Experts: Sinai, Ruelle, Bowen, Alder, Bunimovich, Chernov, Young, Alves, Sarig, etc.

- Coupling method:

- ▶ Hyperbolic systems with nasty discontinuities and singularities, such as dispersing billiards, where BIP is not needed.
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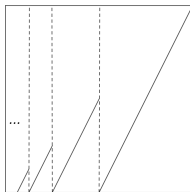
We would like to treat general systems without BIP.

INTUITIVE MODEL I

Consider the piecewise linear map

$$T(x) = \begin{cases} 0, & x = 0, \\ \Lambda_n(x - 2^{-n}), & x \in (2^{-n}, 2^{-n+1}], \quad n \geq 1. \end{cases}$$

where $\inf_n \Lambda_n > 1$.

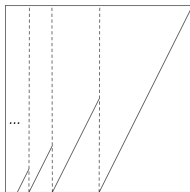


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T does not have BIP, if $\inf_{n \geq 1} 2^{-n} \Lambda_n = 0$.

THEOREM (RYCHLIK, STUDIA. MATH. 1983)

If $\sum_{n \geq 1} \Lambda_n^{-1} < \infty$, then

- T admits an ACIP μ ;
- μ is exponentially mixing and satisfies the CLT.

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- What if $\Lambda_n = n$? Note that $\sum_{n \geq 1} \Lambda_n^{-1} = \sum_{n \geq 1} 1/n = \infty$.

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We shall show $\exists$ ACIP μ .

INTUITIVE MODEL II

van Strien (1996) proposed a linear model for the induced map of the Fibonacci unimodal map: Given any $\lambda \in (0, 1)$, define

$$T_\lambda(x) := \begin{cases} 0, & x = 0, \\ \frac{x - \lambda}{1 - \lambda}, & x \in (\lambda, 1], \\ \frac{x - \lambda^n}{\lambda(1 - \lambda)}, & x \in W_n := (\lambda^n, \lambda^{n-1}] \text{ with } n \geq 2, \end{cases}$$

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THEOREM (BRUIN-TODD, J. LOND. MATH. SOC., 2012)

- if $\lambda \in (0, \frac{1}{2})$, then $\exists$ ACIP with exponential mixing and CLT;
- if $\lambda \in [\frac{1}{2}, 1)$, then no ACIP.

Albeit being linear, the above models have some disadvantages:

- the inverse Jacobian over all branches is not summable, i.e.,

$$\sum_{n \geq 1} \Lambda_n^{-1} = \infty.$$

- the “big image property” fails, i.e., let W_n be the inverse branches,

$$\inf_{n \geq 1} |TW_n| = 0.$$

- typically, the map T can be NOT Markov.

The above disadvantages **prohibited** us to use the functional analytic method of transfer operators.

GENERAL ASSUMPTIONS

We assume

- $T : M = [0, 1] \rightarrow M$ is a Borel measurable map;
- ξ_1 is a countable partition of M into intervals;
- for any $W \in \xi_1$, $T|_{\text{int}(W)}$ is strictly monotonic and C^1 -smooth;
- Moreover,
 - ▶ **(H1)** Chernov's one-step expansion at q -scale;
 - ▶ **(H2)** Log Jacobian is dynamically Hölder wrt ξ_1 ;
 - ▶ **(H3)** Eventual covering over a magnet interval.

(H1) CHERNOV'S ONE-STEP EXPANSION

- Given an interval W , we denote $\{W_\alpha\}_{\alpha \in W/\xi_1}$ the collection of sub-intervals of W after being cut by boundary points of ξ_1 .

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- Given an interval W , we denote $\{W_\alpha\}_{\alpha \in W/\xi_1}$ the collection of sub-intervals of W after being cut by boundary points of ξ_1 .
- Chernov's one-step expansion (at 1-scale):

$$\liminf_{\delta \rightarrow 0} \sup_{W: |W| < \delta} \sum_{\alpha \in W/\xi_1} \frac{|W_\alpha|}{|TW_\alpha|} < 1.$$

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- (H1)** Chernov's one-step expansion at q -scale: $\exists q \in (0, 1]$, s.t.

$$\liminf_{\delta \rightarrow 0} \sup_{W: |W| < \delta} \sum_{\alpha \in W/\xi_1} \left(\frac{|W|}{|TW_\alpha|} \right)^q \frac{|W_\alpha|}{|W|} < 1.$$

Some remarks on the one-step expansion condition:

- if $0 < q \leq q' \leq 1$, then q -scale one-step expansion implies q' -scale one-step expansion;
- N. Chernov and H-K. Zhang (2005, 2009) posed one-step expansion conditions in billiard dynamics: in most case, one-step expansion holds at $q = 1$; the only known case with $q < 1$ is the standard Bunimovich stadia;
- this condition implies “hyperbolicity beats singularities”;
- the previous models satisfies one-step expansion at q -scale for some $q < 1$.

ASSUMPTIONS (H2) AND (H3)

(H2) Regularity of Log Jacobian: $\exists C_J > 0, \gamma_J \in (0, 1)$ such that for any interval $W \in \xi_1$, $\log |T'|$ is (C_J, γ_J) -Hölder continuous on W .

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A priori, we do not have an invariant measure to begin with, we impose the following topological condition:

(H3) Eventual covering: $\exists$ an interval U such that for any interval W , there is $n_W \geq 1$ such that for any $n \geq n_W$, at least one component of $T^n W$ contains U .

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- Such interval U is called “magnet” by Chernov in the billiard case.
- If an ACIP μ is established, then **(H3)** guarantees that μ is mixing.
- **(H3)** is a non-uniform version of “big image property”.
- The previous models have a magnet interval W_2 .

DEFINITION

Let $\mathcal{W} := \{W_\alpha : \alpha \in \mathcal{A}\}$ be a collection of countably many intervals.

A function $f : M \rightarrow \mathbb{R}$ is called a *Hölder series* supported on $\mathcal{W}$ with exponent $\gamma \in (0, 1]$ and power $t \in [0, 1]$, if $f = \sum_{\alpha \in \mathcal{A}} f_\alpha$ such that

- each f_α is supported on W_α , and it is γ -Hölder continuous;
- $\|f\|_{\mathcal{W}, \gamma, t} := \sum_{\alpha \in \mathcal{A}} |W_\alpha|^t \|f_\alpha\|_\gamma < \infty$.

We denote by $\mathcal{H}_{\mathcal{W}, \gamma, t}$ the space of such functions.

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- $\mathcal{H}_{\mathcal{W}, \gamma, t} \subset \mathcal{H}_{\mathcal{W}, \gamma, t'}$ if $0 \leq t \leq t' \leq 1$.
- $\mathcal{H}_{\mathcal{W}, \gamma, 0}$ consists of bounded functions, while $\mathcal{H}_{\mathcal{W}, \gamma, t}$ contains some **unbounded functions when** $t > 0$. For instance, for any $\beta \in (0, 1)$, $f(x) = x^{-\beta} \in \mathcal{H}_{\mathcal{W}, \gamma, t}$ for any $t \in (\beta, 1]$.

RESULT I - EXISTENCE AND DENSITY OF ACIP

From now on, we assume **(H1)-(H3)**.

THEOREM (C-ZHANG-ZHANG, 2019)

- There exists a unique ACIP μ ;
- There are $C > 0$ and $\theta \in (0, 1)$ such that $\|T_*^n Leb - \mu\|_{TV} \leq C\theta^n$;
- The density $h = d\mu/dLeb$ is a Hölder series with exponent γ and power s , for any $\gamma < \gamma_J$ and $s \in (1 - q, 1]$.

RESULT II - EXPONENTIAL MIXING

THEOREM (C-ZHANG-ZHANG, 2019)

Exponential Mixing: For any $t \in [0, q)$, there are $C > 0$ and $\theta \in (0, 1)$ such that for any $f \in \mathcal{H}_{\mathcal{W}, \gamma, t}$ and any $g \in L^\infty(m)$,

$$\left| \int f \cdot g \circ T^n d\mu - \int f d\mu \int g d\mu \right| \leq C\theta^n \|f\|_{\mathcal{W}, \gamma, t} \|g\|_\infty.$$

RESULT III - ASIP AND CLT

THEOREM (C-ZHANG-ZHANG, 2019)

Almost Sure Invariance Principle: Fix any $t \in [0, \frac{1}{2})$. Let $f \in \mathcal{H}_{W,\gamma,t}$ with $\int f d\mu = 0$ such that

- f is adapted;
- f has fast tail;
- f has finite first moment for its autocorrelations.

then the random process $\{f \circ T^n\}_{n \geq 0}$ satisfies ASIP (and thus CLT), i.e.,

$$\left| \sum_{k=0}^{n-1} f \circ T^k - W(n\sigma_f^2) \right| = \mathcal{O}(n^\eta), \quad \text{a.s..}$$

for some $\eta \in (0, \frac{1}{2})$, where $W(\cdot)$ denotes the standard Brownian motion.

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Almost Sure Invariance Principle $\Rightarrow$ Central Limit Theorem

DEFINITION

- A collection $\mathcal{W}$ is *adapted* if every $W \in \mathcal{W}$ belongs $\sigma(\xi_n)$ for some $n(\alpha) \in \mathbb{N}$. A function $f \in \mathcal{H}_{\mathcal{W},t}$ is *adapted* if the collection $\mathcal{W}$ is adapted, and

$$\|f\|_{\mathcal{W},t}^{ad} := \sum_{\alpha \in \mathcal{A}} \mu(W_\alpha)^t \|f_\alpha\|_{W_\alpha, \gamma} < \infty.$$

- Assume that $t \in [0, \frac{1}{2})$. We say $f \in \mathcal{H}_{\mathcal{W},t}^{ad}$ has *fast tail* if there is $a > \max \left\{ \frac{11}{2}, \frac{2+3t}{1-2t} \right\}$ such that

$$\sum_{\alpha \in \mathcal{A}: n(\alpha) \geq n} \|f_\alpha\|_{L^{1/t}(\mu)} = \mathcal{O}(n^{-a}).$$

- the auto-correlations f has *finite first moment*, if

$$\sum_{n=0}^{\infty} n |\text{Cov}_\mu(f, f \circ T^n)| < \infty.$$

STANDARD PAIRS AND STANDARD FAMILIES

Choose $\gamma \in [\gamma_J, 1)$ and $C_r \gg C_J$. To proceed the coupling lemma, Dolgopyat (2000) proposed the following concept.

DEFINITION

- (W, ν) is a *standard pair* if
 - ▶ ν is a probability measure supported on an interval W ;
 - ▶ $\nu \ll \text{Leb}$, and semi-norm $\left| \log \frac{d\nu}{d\text{Leb}} \right|_\gamma \leq C_r$.
- A *standard family* is a convex combinations of standard pairs.

$$\mathcal{G} = \sum_{\alpha \in \mathcal{A}} \lambda_\alpha (W_\alpha, \nu_\alpha),$$

where $\mathcal{A}$ is a countable set, $0 \leq \lambda_\alpha \leq 1$ and $\sum_{\alpha \in \mathcal{A}} \lambda_\alpha = 1$.

TOTAL MEASURE AND CHARACTERISTIC $\mathcal{Z}$ FUNCTION

DEFINITION

Given a standard family $\mathcal{G} = \sum_{\alpha \in \mathcal{A}} \lambda_{\alpha}(W_{\alpha}, \nu_{\alpha})$,

- The total measure of $\mathcal{G}$ is given by

$$\nu_{\mathcal{G}} = \sum_{\alpha \in \mathcal{A}} \lambda_{\alpha} \nu_{\alpha}.$$

- The average length of intervals in $\mathcal{G}$ is measured by the following *Chernov's characteristic $\mathcal{Z}$ function*

$$\mathcal{Z}(\mathcal{G}) := \sum_{\alpha \in \mathcal{A}} \lambda_{\alpha} |W_{\alpha}|^{-q}.$$

Choose $C_p \gg 1$. A standard family $\mathcal{G}$ is *proper* if $\mathcal{Z}(\mathcal{G}) < C_p$.

COUPLING BETWEEN TWO STANDARD FAMILIES

THEOREM (C-ZHANG-ZHANG, 2019)

Coupling Lemma: Under Assumptions (H1)-(H3), there are $C > 0$ and $\theta \in (0, 1)$ such that for any two proper standard families $\mathcal{G}^1$ and $\mathcal{G}^2$,

$$\|T_*^n \nu_{\mathcal{G}^1} - T_*^n \nu_{\mathcal{G}^2}\|_{TV} \leq C\theta^n,$$

where $\|\cdot\|_{TV}$ denotes the total variation norm.

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The existence of ACIP, and exponential decay of correlations are immediate consequences of the coupling lemma.

Thank you very much!